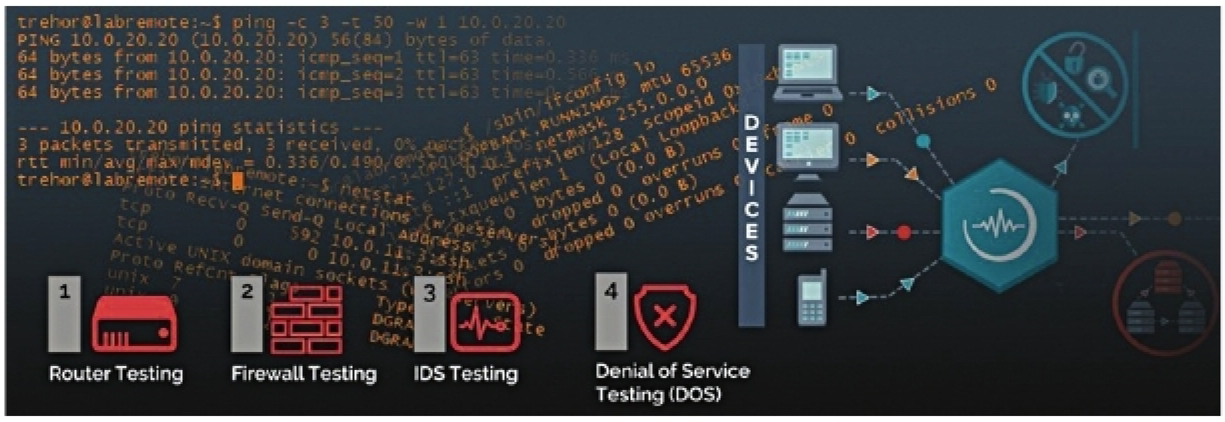
Madhukar Tripathi, head, Optical Business & Marketing, Anritsu India, says, "Testing in wireless networks is entirely different as compared to wired networks. The wireless network is unpredictable due to various radio frequency (RF) transmissions and its harmonic generation for many non-visible, uncontrolled reasons, including illegal transmitters. Such a scenario is often seen in over-the-air (OTA) interference testing, while wired testing is free of such scenarios. OTN analyser or Ethernet analyser or OTDR are some of the popular wired technology testing tools that are used in connected mode with the network."

When applications become complex, testing on every level gains the utmost importance. If the design does not consider variations that exist in real-life applications when compared to a simulation environment, the extra cost is incurred after repairing the faults. Also, it is very common for cybercriminals to redirect traffic on the connected devices to fulfill their malicious intent. Since maintaining network security is an absolute necessity for any application, regardless of its scale, methods for mitigating issues like security breaches due to trojans need to be chosen such that intensive bandwidth requirements are avoided.



Testing procedure for network changes (Image source: *https://ipfabric.io*)

Parameters impacting network performance

There are numerous parameters that must be analysed to quantitatively and qualitatively assess a given network's performance at a time and deal with any bottlenecks that might exist. The goal is to increase network availability and minimise the time for data transfer. A few of them are discussed next.

Packet loss: Packet loss occurs when data transmitted in the form of packets fails to reach its destination. This can be caused by congestion, inadequate infrastructure, software bugs, among others. Although it depends on the application, a loss above two to three per cent is considered significant. It also results in a reduction of the speed or throughput of a network, leading to problems like incomplete transmission of documents. Analysing the root cause and taking measures like updating hardware and software regularly can help reduce the loss.

Latency: Latency, measured in milliseconds (ms), is defined as the amount of time taken for data to travel from one place to another. Lower the latency, better the network performance. There are multiple reasons why the latency in a network may be higher. For example, if one link is experiencing heavy traffic, TCP/IP protocol places packets in a queue for later processing to avoid traffic. Other factors include a large number of routers and a high refractive index of optical fibre. Latency is lower in wired networks.

Bandwidth: The term bandwidth is a theoretical measure of the amount of data that can be possibly transferred over a given time period. It tells us about the capacity of a channel for both wired and wireless networks. It depends on the number of concurrent devices as well as the distance from the router. It is usually measured in bits per second.

Throughput: In telecommunication, throughput is a metric that determines the rate of successful data delivery over a network or communication channel in a pre-defined time frame. Throughput includes the sum of all data rates delivered to individual terminals to calculate speed.

Jitter: Jitter is the variation in time delay or increased latency for the data packets sent over a network, resulting from network congestion and route changes. Jitter is measured in milliseconds (ms) and should be as low as possible to avoid packet loss. Jitter values of a few ms are not even perceptible, but high jitters must be avoided, especially in streaming and gaming.

Testing and measurement

By capturing and analysing network data, problems like slow response time and intrusion can be identified. Key players in this market include Anritsu, Rohde & Schwarz, Fluke Networks, National Instruments, Agilent Technologies, Electro Rent, and so on. The increasing demand for accurate and reliable procedures and tools, especially for testing wireless networks in real time, is pushing the companies to use tools that incorporate remote diagnostic capabilities to figure out the root cause, among other advanced features, for better user experiences.

Tripathi says, "Selection of right testing tools and accessories is very important for any network testing. Testing tools with automation features save time and provide reliable results. There are many tools available for network testing. It depends on the element or network component system under test. Network testing tools must have easy to use Graphical User Interface (GUI) and future technology support proof. Testing tools must have automation tools to support easy, faster, and reliable testing. Automation tools save a lot of time and help deliver quality products in the market. The testing tool must be flexible and scalable for future use."